AZIM PREMJI UNIVERSITY, BHOPAL
First Semester 2026-2027
Tutorial Sheet-4

Course No. MTH-113 Course title: Introduction to Quantitative Thinking

---

(1) Point $(-3, 5)$ lies in the

(a) first quadrant (b) second quadrant
(c) third quadrant (d) fourth quadrant

(2) Signs of the abscissa and ordinate of a point in the second quadrant are respectively

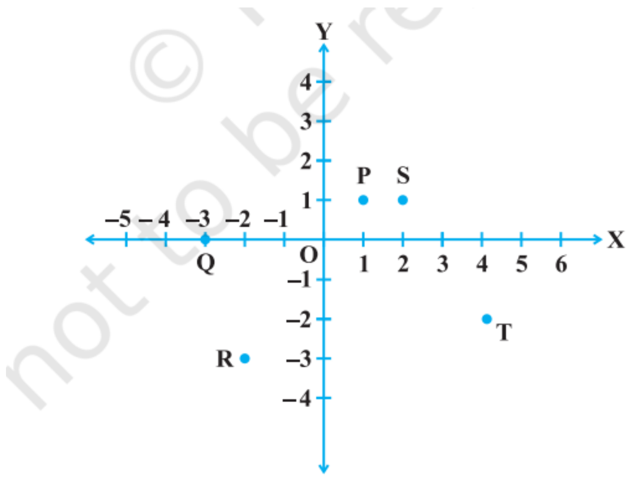
(a) $+, +$ (b) $-, -$
(c) $-, +$ (d) $+, -$

(3) Point $(0, -7)$ lies

(a) on the x-axis (b) in the second quadrant
(c) on the y-axis (d) in the fourth quadrant

(4) Find the perpendicular distance of the point $P(3, 4)$ from the y-axis.

(5) Write the coordinates of each of the points P, Q, R, S, T and O from the following figure.



(6) Plot the points (x, y) given by the following table:

x	2	4	-3	-2	3	0
y	4	2	0	5	-3	0

(7) Points $A(5, 3)$, $B(-2, 3)$ and $D(5, -4)$ are three vertices of a square ABCD. Plot these points on a graph paper and hence find the coordinates of the vertex C.

(8) The graph of the linear equation $2x + 3y = 6$ cuts the y-axis at the point:

(9) Any point on the line $y = x$ is of the form:
(a) (a, a) (b) $(0, a)$ (c) $(a, 0)$ (d) $(a, -a)$

(10) The point of the form $(a, -a)$ always lies on the line

(a) $x = a$ (b) $y = -a$ (c) $y = x$ (d) $x + y = 0$

(11) Draw the line passing through $(5, 6)$ and $(-2, -1)$, and hence find another point from this line.

(12) Draw the line $y = 2x + 3$ in the graph, and find the points where the line touches the x-axis and y-axis.

(13) 8 more than three times the number x can be represented as:
(a) $8 + x + 3$ (b) $3x - 8$
(c) $3x + 8$ (d) $8x + 3$

(14) 7 times of y subtracted from 50 can be expressed as:

(15) Translate each of the following statements into an equation, using x as the variable:
(a) 13 subtracted from twice a number gives 3.
(b) One fifth of a number is 5 less than that number.
(c) Two-third of number is 12.
(d) 9 added to twice a number gives 13.
(e) 1 subtracted from one-third of a number gives 1.

(16) Arjun bought a rectangular plot with length x and breadth y and then sold a triangular part of it whose base is y and height is z. Find the area of the remaining part of the plot.

(17) Amisha has a square plot of side m and another triangular plot with base and height each equal to m. What is the total area of both plots?

(18) Shiv works in a mall and gets paid Rs 200 per hour. Last week he worked for 7 hours and this week he will work for x hours. Write an algebraic expression for the money paid to him for both the weeks.

****